3 Food and Health

We need food to stay alive. It is essential to eat a balanced diet for the proper growth and development of our body. We will learn about the importance of eating a balanced diet, the harmful effects of junk food, and food adulteration in this chapter. Before learning them, let us first revisit the concept of components of food.

Learn about

- Components of food
- Balanced diet
- Junk food
- Ways to make diet healthier
- Diseases caused by incorrect food habits and unhealthy lifestyle
- Prevention of lifestyle diseases
- Deficiency diseases
- Food adulteration

Components of Food

Food is necessary for the survival of living beings. It gives us energy and it is necessary for growth and repair of the body. There are five main components of food: carbohydrates, fats, proteins, vitamins, and minerals. These components are also known as **nutrients**.

The table given below lists the functions and sources of these components of food.

Components of food	Function	Sources
Carbohydrates	Provide energy to the body	Cereals (e.g., rice, wheat, and corn), fruits, sugar, potato
Fats	Provide energy to the body	Oil, milk, butter, cheese, ghee, cream, dry fruits
Proteins	Essential for growth and repair of the body and muscle-building	Eggs, meat, fish, milk, cheese, soya bean, pulses
Vitamins	Needed in small amounts for the normal functioning of the body and to prevent diseases	Fruits, green leafy vegetables, fish, eggs
Minerals (iron, calcium, iodine, phosphorus, etc.)	Needed in small amounts for keeping the body healthy	Vegetables, fruits, milk, eggs

In addition to these five components, water and roughage are also essential to our diet.

Water We need adequate amount of water so that our body stays healthy. It helps in the transportation of various substances like oxygen, nutrients, etc.

Roughage It is the undigested portion of the plant food, which includes mostly fibres. It is largely present in whole grains and in various fruits and vegetables. It helps in the easy passage of food through our digestive system and prevents constipation.

▶ Balanced Diet

We need all the components of food in adequate amounts to remain healthy. *A diet that contains all the components of food in the right proportions, for the healthy functioning of our body is called a* ***balanced diet***.

Components of a Balanced Diet

Eating a balanced diet provides the required nourishment to the body and keeps it healthy. The different components that need to be a part of a balanced diet are carbohydrates, proteins, fats and sugars, milk and dairy products, and fruits and vegetables. The amount of each of these components to be included in our diet generally depends on the (a) age and (b) the amount of physical activity a person does.

Fact File

Rice is the main food for nearly half of the world's population because it is high in fibre, nutritious, inexpensive, and can be stored for a long period.

Look at the following examples:

- A person who does more physical labour (such as an athlete or a labourer) needs a diet that is richer in carbohydrates than a person who works in an office.
- Growing children need to have a protein-rich diet to help in the overall development of their bodies and minds.
- Adults who have jobs that require them to sit and work during the day, need a diet that has less carbohydrates and fats to prevent them from gaining weight.

A balanced diet contains different components of food in adequate amounts

- Elderly persons usually need a low-fat, high-fibre diet along with moderate exercise. They should have plenty of milk and milk products, fresh fruits, and vegetables to meet their vitamin and mineral needs.

Importance of Eating a Balanced Diet

It is very important to eat a balanced diet as it promotes the following:

- **Growth and development of the body** A balanced diet is especially important for children as it helps in the proper development of their growing bodies. A balanced diet also provides energy to all the body parts so that they can perform their functions properly.
- **Growth and development of mind** A balanced diet helps in improving memory, concentration, and allows us to focus better on our day-to-day activities.
- **Maintains an ideal body weight** A balanced diet helps in maintaining an ideal body weight. It helps in preventing weight gain.
- **Prevents diseases** A balanced diet gives our body the strength to fight infections and disease-causing organisms. It also prevents diseases caused due to lack of nutrients.

REMEMBERING UNDERSTANDING

Questions

Match the following.

1. Carbohydrate	a. Growth and repair of body
2. Vitamins	b. Provides energy
3. Roughage	c. Burgers, pizzas, patties
4. Protein	d. Prevents diseases
5. Junk Food	e. Undigested portion of food

▶ Junk Food

*The term **'junk food'** refers to any food that contains high amount of sugar and fat, but low amount of other nutrients and fibres.* When eaten, these food items make us feel full, but they provide less nutrition to the body. All the excess fat present in these foods cannot be used completely by the body and gets stored. This leads to weight gain and poor health over a period of time. Examples of junk food include packaged foods such as salted potato chips, candies, chocolates, ice creams, and sugary cold drinks, and other fatty foods such as patties, burgers, pizzas, samosas, and sugary sweets.

Adverse Effects of Eating Junk Food

We should not make junk food a part of our daily diet. Eating junk food regularly can have several adverse[1] effects on our health and well-being. Some common ill effects of eating junk food are: Deficiency[2] of certain nutrients, weight gain and obesity, lack of energy, poor digestion, high levels of salt and sugar in the blood that leads to diseases such as high blood pressure, heart diseases and diabetes, respectively.

▶ Ways to Make Diet Healthier

We should avoid junk food and eat a balanced diet. There are many ways of making the diet healthier. Some of them are as follows:

- Use less sugar, salt, and oil while cooking food.
- Avoid deep frying and reusing oil.
- Eat fruits like apples, oranges, and guavas daily for vitamins.
- Add **sprouts** of gram and *moong dal* to at least one meal in a day. Eat lots of green vegetables.
- Add fibre to your diet. Use whole grains instead of polished cereals.
- Eat lots of salad and yogurt.
- Eat **fermented food** regularly. It contains many useful bacteria that help in digestion. Some common fermented food items are *idli*, *vada*, *dosa* (South India), *dhokla* (Gujarat), and *panta bhaat* (East India).

Know Your SDGs

SDG 3: **Good Health and Well-Being**

Maintaining a healthy diet is crucial for overall well-being. How can you reduce the use of unhealthy ingredients like sugar, salt, and oil in your meals?

Activity

EXPERIENTIAL LEARNING · LIFE SKILLS · 21st CENTURY SKILLS

Aim: To classify food items into junk food and healthy food

Materials required: Chart paper, glue, a marker pen, pictures of various food items (e.g., milk, chips, fruits, vegetables, etc.) from magazines and newspapers (or draw your own)

Fresh fruits and vegetables (healthy food)

[1]**adverse:** harmful
[2]**deficiency:** lack or shortage

Procedure:

1. Using the pen, divide the chart paper into two columns—A and B.
2. Write 'Healthy Food' as the heading in Column A and 'Junk Food' as the heading in Column B.
3. Paste the picture of each food item into the correct column. For all the healthy food items, mention the different nutrients they contain.
4. Similarly, identify the junk food items and note down the nutrients present in them.

Observation: Healthy food items contain a variety of nutrients. Junk food items contain large amounts of sugar, salt, and fat, and low amounts of other nutrients.

Conclusion: It is important to include as many healthy food items in your diet and avoid junk food.

Diseases Caused by Wrong Food Habits and Unhealthy Lifestyle

Bad food habits (such as eating a lot of junk food) and an unhealthy lifestyle (such as less or no physical activity) may lead to several diseases. These are called **lifestyle diseases**. Some common lifestyle diseases are obesity, high blood pressure (or hypertension), diabetes, anaemia, and various heart diseases.

Obesity

Obesity *is defined as abnormal or excessive fat accumulation that increases the risk of health problems.*

A child suffering from obesity

Obesity or excessive weight gain is increasingly becoming a common problem among children and adults.

Causes

Weight gain occurs due to eating large amounts of junk food, and also because of lack of physical exercise. People start gaining weight if they regularly eat junk food, or mostly stay indoors (in offices or homes) doing activities that do not involve much physical effort. Over a period of time, this leads to excessive weight gain or obesity.

Symptoms

- The first sign of obesity is an unhealthy weight gain. The excess fats from junk food get stored in the body causing an increase in weight. Fat around the waist is considered very bad. It may lead to heart diseases.

- Tiredness and lack of energy are other signs of obesity and poor health. People suffering from obesity become slow in all activities and get tired easily.

High Blood Pressure

High blood pressure (or hypertension) is another common lifestyle disease. High blood pressure is when the pressure in your blood vessels becomes too high. It is common, but can become serious if not treated. It may lead to several heart diseases.

A woman suffering from hypertension

Causes

The causes of hypertension include lack of physical exercise, stress, anxiety, and intake of food rich in salt and fat.

Symptoms

- In some people, it may cause heaviness in head and dizziness.
- It may also cause breathlessness or excessive sweating on slight physical activities.

Diabetes

Diabetes is a condition in which the sugar levels in your blood become too high. Nowadays, people lead sedentary[3] lives, avoiding walking and physical activity. They often eat junk food rich in carbohydrates and fats. This leads to obesity, which can sometimes cause diabetes.

A person suffering from diabetes

Causes

It is caused due to obesity, lack of exercise, and stress.

Symptoms

- Patients suffering from diabetes suffer from excessive thirst, excessive hunger, and excessive passing of urine.
- They also suffer from urinary, skin, and lung infections.
- In some people, diabetes may also lead to high blood pressure, tuberculosis, and heart diseases.

Think and Discuss 21st CENTURY SKILLS

Not everyone who is overweight or obese gets diabetes. What other factors might play a role?

[3]**sedentary:** involving little to no physical activity

Anaemia

Anaemia is when you have a deficiency of red blood cells in the blood, resulting in pale skin and tiredness. It is caused due to, malnutrition, blood loss, and worm infestation[4].

A person suffering from anaemia

Causes

It is caused due to lack of iron in our diet. Eating junk food rich in sugar, salt, and fat, and avoiding healthy food rich in iron (e.g., spinach and lentils) may also lead to anaemia.

Symptoms

- Fatigue (tiredness) and weakness
- Breathlessness
- Loss of appetite
- Pale skin
- Anaemia may also lower the body's ability to fight infections and disease-causing organisms.

▶ Prevention of Lifestyle Diseases

We can prevent lifestyle diseases by adopting a healthy and stress-free lifestyle. Some ways to prevent it are given below.

- Eat a balanced diet that contains important nutrients. Include fresh fruits and green vegetables in the diet. Avoid eating junk food. Stay away from foods that contain excess of salt or sugar.
- Exercise regularly. Practice yoga under the guidance of an expert or in the presence of a teacher/adult. Also, spend time outdoors and do activities such as walking, running, swimming, and cycling. Playing games such as football, cricket, tennis, badminton, and so on is also a good exercise.
- Avoid spending too much time indoors. Watch television or play games on computers, tablets, or smartphones for short intervals of time only.
- Set a healthy sleeping routine for every day. Get up early in the morning and go to bed early at night.
- Lead an active life. Avoid using vehicles for short journeys. Instead, walk or cycle, if possible. Use stairs instead of lifts or escalators, wherever we can.
- Avoid sitting or lying down for long period of time. Take breaks and move around at regular intervals.

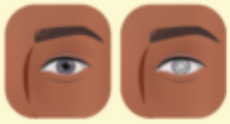
[4]**worm infestation:** presence of worms in stomach and intestine

- Drink at least 8 glasses of water every day.

Some healthy lifestyle habits

Deficiency Diseases

Regular lack or deficiency of a particular nutrient in the diet may cause certain diseases. Such diseases are called deficiency diseases. Some common deficiency diseases are given in the following table.

Deficiency disease	Symptoms	Causes	Prevention
Kwashiorkor	Swelling in the abdomen and other parts of the body due to accumulation[5] of fluids, flaky and dry skin, reduced growth, frequent infections	Deficiency of proteins	Eat diet rich in proteins, carbohydrates, and fats.
Marasmus	Significant weight loss, loose and dry skin folds, fluid accumulation, and diarrhoea (excessive loose motions)	Deficiency of proteins and carbohydrates	Eat diet that includes milk, meat, eggs, and cereals.
Night blindness  Before After	Inability to see at night or in poor light	Lack of vitamin A in the diet	Eat diet that contains green leafy vegetables, papayas, carrots, mangoes, milk, butter, and egg yolk.

[5]**accumulation:** storage

Anaemia	Feeling of weakness or fatigue	Lack of iron in the diet	Eat lots of green leafy vegetables such as spinach, fruits such as apples and guavas, jaggery, and meat.
Rickets	Softening of bones, bowlegs, and knock-knees	Lack of vitamin D in the diet	Eat a diet with milk, dairy products, and saltwater fish. Spend some time in the sun daily, especially in the morning, to help form vitamin D.
Scurvy	Swollen gums that bleed easily	Lack of vitamin C in the diet	Regularly eat citrus fruits such as amla, oranges, lemons, and tomatoes.
Beriberi	Fatigue, loss of sensation in hands & feet, and weakness	Lack of vitamin B1 in the diet	Eat diet that includes milk, peas, cereals, and green vegetables.
Goitre	Swelling of the front portion of neck	Lack of iodine in the diet	Use iodized salt for cooking.

▶ Food Adulteration

Food adulteration *refers to the addition of unwanted and unsafe substances to food items. The unwanted substances that are added to a food item, to increase its quantity and presentation are called* ***adulterants***. These added substances may adversely affect the health of the buyers when the adulterated food item is consumed.

"Why are food items sometimes adulterated by sellers?"

- Sellers may add other food items which are of less value to increase the quantity and make more profit.

- They may also add colours to a food item to make it look more attractive to the buyers.

Food adulteration is mainly done to cheat buyers and make profit. It is a punishable act by law in India.

Fact File

In India, the Food Safety and Standards Authority of India (FSSAI) keeps a check on food adulteration. Anyone starting a food business must obtain an FSSAI license.

Common Examples of Food Adulteration

- A commonly seen food adulteration is the addition of water to milk by milk sellers. This is done to increase the volume of milk. The milk seller thus makes a profit by selling this adulterated milk.
- Other common adulterants are chalk powder, marble chips, stone chips, and harmful colours. These are often found in cereals and pulses.
- Common kitchen spices such as turmeric powder and chilli powder are often adulterated with lead pigments.

Case Study

Rajesh, a shop owner in a village, sells different types of spices. Recently, he has noticed that some of his competitors have started adding sawdust, brick powder, or synthetic powder to spices to make them look more colourful. Rajesh knows this is wrong, but he's worried about losing customers to these competitors.

Based on the above information, answer the following questions:

a. What substances are being added to the spices by Rajesh's competitors?

b. What could happen if people eat these adulterated products?

Wrap Up

- There are five main components of food: carbohydrates, fats, proteins, vitamins, and minerals. These components are also known as nutrients.
- A diet that contains all the components of food in the right proportions for the healthy functioning of our body is called a balanced diet.
- It is very important to eat a balanced diet as it promotes growth and development of the body and mind, helps in maintaining an ideal body weight, and also prevents diseases.
- The term 'junk food' refers to any food that contains high amount of sugars and fats, and low amount of nutrients and fibres. Regular eating of junk food can have several adverse effects on our health and well-being.

- Bad food habits (eating a lot of junk food) and an unhealthy lifestyle (little to no physical activity) may lead to lifestyle diseases like obesity, high blood pressure (or hypertension), diabetes, anaemia, and various heart diseases.
- Some diseases are caused due to the lack or deficiency of a particular nutrient in the diet. These are called deficiency diseases.
- Food adulteration refers to the addition of unwanted and unsafe substances to food items.

Exercises

SECTION I

A Choose the correct option.

1. Which of these is the main component of food?
   a. Fats b. Carbohydrates c. Proteins d. All of these
2. Oil, milk, butter, and cheese mainly contain .....
   a. roughage b. fats c. proteins d. vitamins
3. Which among these is not a lifestyle disease?
   a. Diabetes b. Anaemia c. Scurvy d. Obesity
4. Rickets is caused due to the lack of .....
   a. vitamin A b. vitamin B c. vitamin C d. vitamin D
5. Goitre occurs due to the lack of .....
   a. calcium b. iron c. iodine d. protein
6. Which among these is not an adulterant?
   a. Milk b. Chalk powder c. Marble chips d. None of these

B Assertion and Reasoning questions

1. **Assertion (A):** Food adulteration is done to improve the nutritional value of food.

   Reason (R): Food adulteration involves adding unwanted and unsafe substances.

   a. Both A and R are True b. Both A and R are False
   c. A is True and R is False d. A is False and R is True

2. **Assertion (A):** Eating a lot of fruits and vegetables can prevent deficiency diseases.

   Reason (R): Fruits and vegetables are rich in vitamins and minerals.

   a. Both A and R are True
   b. Both A and R are False
   c. A is True and R is False
   d. A is False and R is True

C Choose the correct option to fill in the blank.

1. There are ..... (four/five) main components of food.
2. Minerals are needed in ..... (large/small) amounts by our body.
3. Roughage includes mostly ..... (water/fibres).
4. Junk food contains high amount of ..... (sugar/roughage).
5. It is ..... (important /not important) to eat a balanced diet.

D Give two examples of the following.

1. Vitamin-rich foods ....
2. Carbohydrate-rich foods ....
3. Junk foods ....
4. Fermented foods ....

SECTION II

E Short answer questions

1. Why should roughage be present in our diet?
2. How does a balanced diet help in the growth and development of the mind?
3. Write any three harmful effects of junk food.
4. How is anaemia caused?
5. What are the symptoms of night blindness?

F Long answer questions

1. You're packing a lunch for a long hike. Considering the importance of a balanced diet, what kind of foods would you include to ensure you have enough energy for the entire hike?
2. You're helping your family switch to a healthier lifestyle. List three specific ways you could modify your family's diet to make it healthier.
3. Differentiate between lifestyle diseases and deficiency diseases.
4. Write the symptoms, causes, and prevention of any two deficiency diseases.
5. Give common examples of food adulteration.

Picture Study

Look at the given picture and answer the questions given below.

1. Write the names of the food items shown, inside the left side boxes; while in right side boxes, write the names of the main nutrients present in them.
2. What does the given picture depict?

   .....

   .....
3. What is the importance of eating such a diet that contains all the nutrients in adequate amount?

   .....

   .....

My Learning Corner

A Think about TP

1. Amit and his grandfather were having a tea-time conversation. Grandfather mentioned the importance of a healthy diet and avoiding junk food. Amit agreed, noting his Science teacher had said the same. Can you suggest some other ways to make our diet healthier?

   .....

   .....
2. Pranab is suffering from swollen and bleeding gums. His doctor has suggested to his parents to give him citrus fruits regularly. Can you suggest why?

   .....

   .....

B Try out

1. Collect pictures of food items and paste them in your Science scrapbook under the headings—'Healthy Food' and 'Junk Food'. Firstly, give the definitions of healthy and junk food below.

   Healthy Food: .....

   Junk Food: .....

2. Write down the names of food items that you ate last week. Count their total number and identify the number of junk food items among them. Calculate the percentage of junk food using the formula: (Total number of junk food items eaten/Total number of food items eaten) × 100.
   [Remember, it is best to keep the percentage of junk food as low as possible for a healthy body and heart.]
3. Write a slogan on "Junk food and its effects". TP
4. Conduct a group discussion in the classroom on the topic—"Practical solutions on how to avoid junk food". TP
5. Make a report on different kinds of food adulterants that are added to our food items. You can take help of teachers, elders, and the Internet for it. Also, discuss the ill consequences.

Self-Assessment

Now that you have completed the chapter, score each of the following tasks from 1 to 5 to indicate how well you can do them.

Score 5 = I can definitely do this.

Score 1 = I cannot do this yet.

I can...	My score
• discuss the components of a healthy diet.	
• explain the importance of a balanced diet.	
• list healthy and junk food items and differentiate between them.	
• suggest/find out some ways to make diet healthier.	
• identify symptoms and prevention methods for lifestyle diseases like obesity, anemia, diabetes, and high blood pressure.	
• discuss causes and prevention of deficiency diseases.	
• raise awareness regarding food adulteration.	

Worksheet

Can you locate the names of 6 lifestyle and deficiency diseases in the word grid given below?

Diabetes	Goitre	Rickets
Hypertension	Anaemia	Scurvy

Z	Q	S	E	H	G	R	B	X	W
O	N	A	S	Y	O	E	B	N	R
V	Y	N	E	P	I	N	O	A	V
D	I	A	B	E	T	E	S	W	T
Y	R	E	U	R	R	O	C	Z	U
U	W	M	E	T	E	T	U	U	O
N	O	I	Q	E	B	M	R	U	M
O	S	A	U	N	V	W	V	L	W
E	M	S	Q	S	C	Z	Y	E	Z
M	O	H	R	I	C	K	E	T	S
W	Q	T	P	O	W	I	C	Z	E
M	G	L	F	N	Q	N	Y	U	B